

Part I

SUMMARY of Chain Rule

Summary

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- **The Chain Rule:** If $f(x)$ and $g(x)$ are differentiable functions, then $\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$.
- **The Derivatives of Trigonometric Functions:**

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\frac{d}{dx} \cos(x) = -\sin(x)$$

$$\frac{d}{dx} \tan(x) = \sec^2(x)$$

$$\frac{d}{dx} \cot(x) = -\csc^2(x)$$

$$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

$$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x)$$

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 For the following problems, the derivative is given. Determine which function was the original function.

(a) The derivative is $f'(x) = \cos(x)e^{\sin(x)}$. Which is the original function?

(i) $f(x) = (\sin(x))(e^x)$

(ii) $f(x) = \sin(e^x)$

(iii) $f(x) = e^{\sin(x)}$

(iv) $f(x) = e^{x \sin(x)}$

(b) The derivative is $g'(x) = 4(\tan(x^4 - 5x))^3 \sec^2(x^4 - 5x)(4x^3 - 5)$. Which is the original function?

(i) $g(x) = (\tan(x) - 5x)^4$

(ii) $g(x) = \tan^4(x) - 5x^4$

(iii) $g(x) = \tan(x^4 - 5x)$

(iv) $g(x) = \tan^4(x^4 - 5x)$

Problem 2

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Problem 2

Problem 2 A table of values for $f(x)$ and $f'(x)$ is shown below:

x	$f(x)$	$f'(x)$
1	3	4
2	2	3
3	4	5
4	6	3

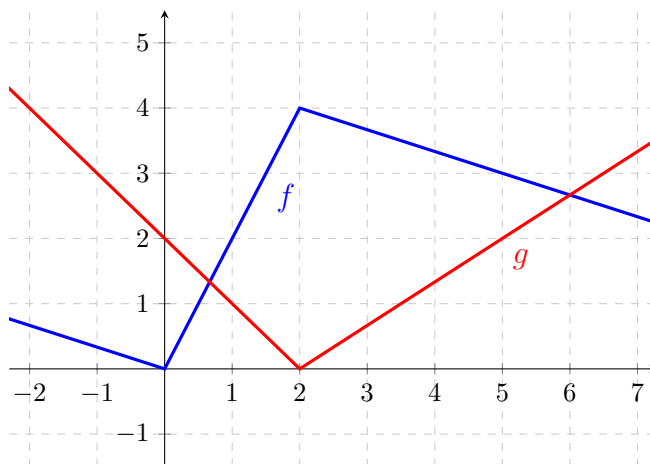
- Evaluate the limit $\lim_{x \rightarrow 2} \frac{f(x^2) - 6}{x - 2}$. **EXPLAIN**.
- Evaluate $\frac{d}{dx} f(f(x))$ at $x = 3$.
 - (a) 6
 - (b) 25
 - (c) 5
 - (d) 15
 - (e) DNE
 - (f) None of the previous answers.

Problem 3

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Problem 3

Problem 3 Given the following graphs of f and g (both piecewise linear functions), define new functions $u(x) = f(g(x))$ and $v(x) = f(x)g(x)$. Find:



(a) $u'(1)$

(b) $v'(1)$

(c) $\lim_{x \rightarrow 1} \frac{\sqrt{g(x)} - 1}{x - 1}$

Problem 4

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Problem 4

Problem 4 Suppose the line tangent to the graph of f at $x = 1$ is $y = 6x - 7$. Find an equation of the line tangent to the following curves at $x = 1$:

(a) $y = g(x) = 5(f(x))^4$

(b) $y = h(x) = x^2(f(x^3))$

Problem 5

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Problem 5

Problem 5 *Differentiate each function (with respect to x)*

(a) $\cos(\sqrt{x+7})$

(b) $\sqrt{\cos(x)+7}$

(c) $\sqrt{\cos(x)+7}$

(d) $\cos(\sqrt{x}+7)$

(e) $\cos(x) \cdot (\sqrt{x}+7)$

Problem 6

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Problem 6

Problem 6 Find the derivative of the following functions:

(a) $f(x) = \sin(x) \cos(x)$

(b) $f(x) = \frac{e^x \tan(x)}{\sec(x) + 2}$

(c) $f(x) = e^{x \tan(x)}$

(d) $f(x) = \sin(x) \cos(x) e^{3x}$

(e) $f(x) = \frac{x + 5}{7x^6 + \cot(x)}$

(f) $f(x) = \sin(2x) \sec^3(x^2 + 4x)$

Problem 7

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Problem 7

Problem 7 Find values for a , b , and c so that the following function is differentiable everywhere.

$$f(x) = \begin{cases} a \sin(x) + b \cos(x) & \text{if } x < 0 \\ ax^2 + bx + c & \text{if } x \geq 0 \end{cases}$$